

Industry Analysis of the Semiconductor Equipment Industry

(with a focus on lithography markets and advanced node manufacturing)

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Industry Analysis

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Industry Overview

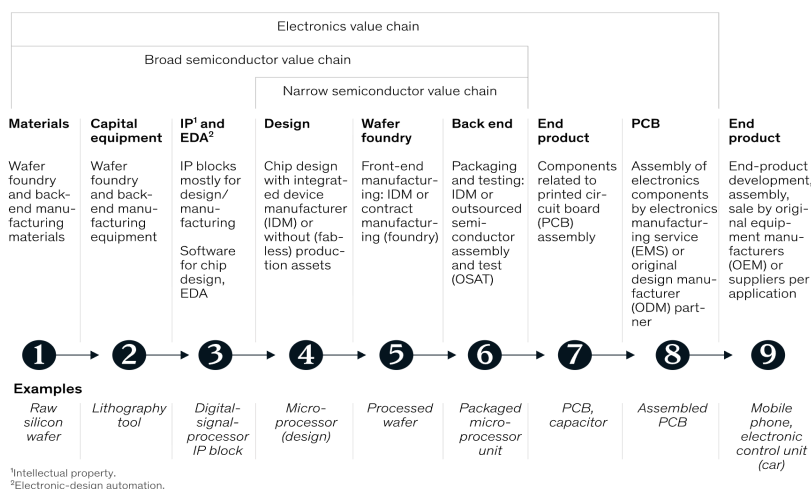
The semiconductor equipment industry produces high precision tools and systems for the manufacturing of integrated circuits. The tools deal with processes such as lithography, deposition, metrology, etching and packaging, all of which are highly important for the chip's performance and in deciding its cost. The industry sits close to the top of the semiconductor value chain, and enables foundries like TSMC, Samsung, and Intel to fabricate chips at smaller and smaller node sizes. Improvements in technologies are pivotal in this industry and hence this segment is strategically critical for the global electronics industry.

The value chain is described to the right, with lithography, deposition, and etch tool suppliers being the enablers for the wafer fabrication process. Lithography determines the minimum feature size achievable on a silicon wafer and hence is a key bottleneck for advanced integrated chip production. Lithography tools are the most expensive, advanced and strategically important tools among all equipment categories.

The present players in the lithography tool segment are ASML, Canon and Nikon. ASML dominates in this segment being the exclusive global supplier of extreme ultraviolet (EUV) systems and is the leader in high-end deep ultraviolet (DUV) systems, with Canon and Nikon participating in the legacy DUV tools sub-market. As advanced integrated chip making moves towards 2nm and beyond, lithography becomes the rate-limiting step in scaling, and this puts ASML at a very advantageous position.

The semiconductor value chain extends from material procurement to back-end manufacturing.

Value chain areas

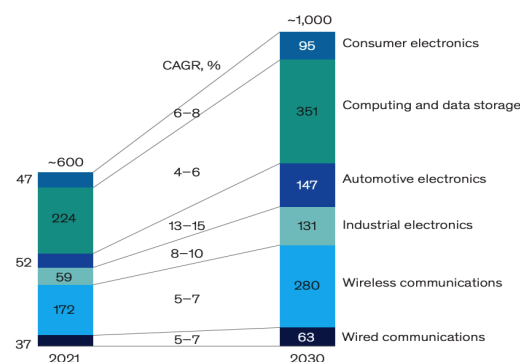


Market Size & Growth

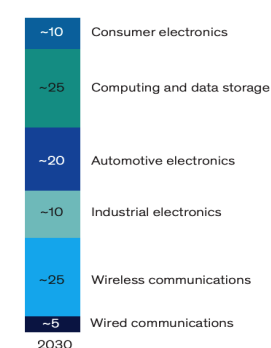
As shown, the global semiconductor market is scaling at an unprecedented pace, expecting to reach \$1 trillion in value by 2030, driven by a booming demand by AI infrastructure, cloud computing, and in other domains. Without proportional investments in the wafer fabrication tools, the industry risks encountering structural bottlenecks that constrain output and delay next-generation node advancements. A robust equipment supply chain is necessary and essential to sustain Moore's law and support the multi-trillion dollar economy that depends on advanced chip manufacturing.

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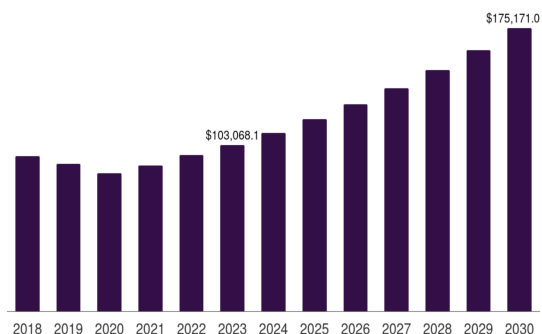
Global semiconductor market, \$ billion



Growth contribution per vertical, %



Global semiconductor manufacturing equipment market size, 2018-2030 (US\$M)

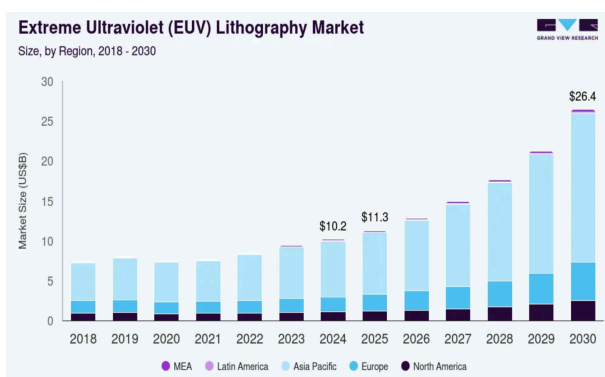


As shown, similar growth in the semiconductor equipment manufacturing industry is very much present and it shows that semiconductor fabs like TSMC, SK Hynix, Samsung need more and better wafer fabrication equipment at an ever increasing pace. The market size of the global semiconductor manufacturing equipment market will almost double between 2023 and 2030, with a CAGR of ~7.8%, matching and sometimes beating the CAGRs in the individual semiconductor market segments shown above, and this makes sense because the semiconductor equipment industry tends to grow at a faster rate than the semiconductor industry itself because advanced systems require a disproportionately larger amount of CAPEX by foundries.

Even more specifically in the EUV lithography market, the market value is projected to skyrocket with a CAGR of ~21.6%, with the year-over-year growth rate not plummeting drastically and staying within a healthy range. This implies strong and steady demand for the lithography market as it shows that foundries are going to demand more and more from the chip manufacturing equipment market over the next decade.

The increase in demand in the overall semiconductor market and the semiconductor manufacturing equipment market in specific are due to the following major megatrends:

- Surge in demand for AI-related memory, in both DDR and HBM, along with advanced logic chips.
- Continuing interconnection between wireless communication, telecom, and cloud technology via connected devices is driving global demand for advanced semiconductor devices.
- Growing populations, urbanization, the energy transition and electrification to support smart mobility are increasing demand for advanced electronic devices.



Industry Structure

The semiconductor manufacturing equipment industry is characterized by being highly concentrated in the hands of a few key players. Throughout lithography, deposition, metrology, etching and packaging each segment has a key resembling feature, that being each segment has 1 or 2 companies that dominate them followed by a large number of small players. For the EUV lithography equipment market in specific the competitive landscape is described below:

Force	Strength	Reasons
Rivalry	Low	Few competitors with virtually zero market share
Supplier Power	High	Several vital elements that only have one global supplier, a key example being Carl Zeiss SMT, who is the sole manufacturer of EUV mirrors and optical assemblies.
Customer Power	High	Few large buyers, like TSMC, Samsung, and Intel.
Substitutes	Virtually None	No alternative to EUV as of present.
Entry Barriers	Very High	Requires high capital and precision engineering, protected by multiple patents.

While it's known that the electronics industry has a very high barrier to entry, the lithography industry takes it to an extreme, with the barriers to entry being the highest of any industry in the world. Driven by extreme technological complexity, large CAPEX requirements, highly loyal suppliers, intellectual property protection, decades of customer trust, and being treated as national security technology by governments, makes market entry for new competitors virtually impossible.

Key Players

Company	Segment	Strength
ASML	Lithography (EUV and DUV)	Monopoly in EUV, strategic bottleneck.
Applied Materials	Deposition	Broad product range, largest by revenue.
Lam Research	Etching	High margins, strong foundry loyalty.
Tokyo Electron	Coating	Key player in Asian markets.

Regulations & Geopolitical Impact

U.S. Driven Export Controls:

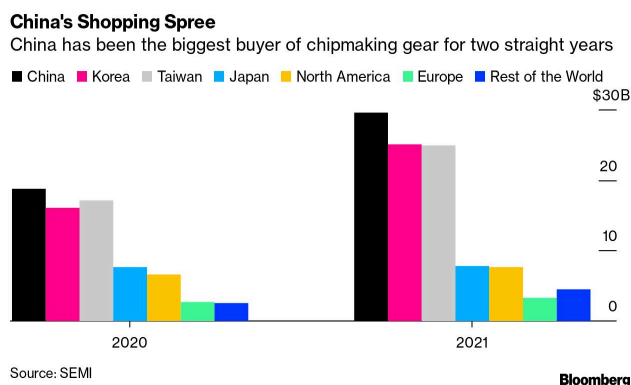
- In 2022 the U.S.'s Department of Commerce's Bureau of Industry and Security (BIS) laid down a strict set of import controls on Russia's access to technologies during the Russo-Ukrainian War, which heavily impacted Russia's access to semiconductors and semiconductor equipment.
- In 2024, the U.S. announced that they have imposed a new set of export controls on China, restricting the sale of semiconductor manufacturing equipment and high-bandwidth computer memory.

Dutch Government Restrictions:

- In 2024, the Dutch government announced an updated license requirement regarding the export of immersion DUV semiconductor equipment, causing major lithography equipment companies to apply for new export licenses.
- Under U.S. pressure, the Dutch government also applied restrictions on the export of High-NA (Numerical Aperture) EUV lithography systems, to mainly Chinese businesses.

China's Role:

- China has and still is the largest buyer of advanced node tools, driven by stockpiling before sanctions.
- The Chinese government has been pushing for a domestic manufacturer of lithography systems, particularly SMEE (Shanghai Micro Electronics Equipment), though it's unknown whether they can achieve this within the desired timeline.



Global Subsidies:

- U.S. CHIPS and Science Act: Passed in 2022, this act intends to bring back microchip manufacturers back to the U.S.A. after decades of offshoring the technology.
- Subsidies from the E.U. : Similar to the U.S.'s Chips Act, the E.U. Chips Act aims to strengthen the E.U.'s semiconductor ecosystem.
- Incentives from Japan, Korea and Taiwan: Japan and Taiwan have collaborated through the JASM venture, and South Korea has announced support packages for the chip industry in 2024.

National Security importance of AI Hardware:

AI hardware has become of central importance to national security all around the globe, because military capability, intelligence, and technology are all determined by A.I. computational power. As a result governments are aggressively attempting to regulate semiconductor supply globally and are also investing billions into developing domestic semiconductor manufacturing to maintain a strategic high ground in the war for AI control.

Industry Outlook

In the long term the semiconductor manufacturing equipment industry is strategically at a very strong position right now, with A.I. hyper-scaling being the main cause for the exponentially increasing demand in semiconductor equipment and the large global race for achieving supremacy in the 'Chips Race' (a term I've coined) also accelerating demand for not only lithography technology but in the whole semiconductor manufacturing equipment industry in general. If the A.I. bubble hypothesis were to be true this industry would suffer severely as a surge in CAPEX in chip manufacturing equipment are driven by rapid AI driven funding for AI microchips.

Though the industry does remain sensitive to cyclical changes in demand and geopolitical restrictions, it's no doubt that, at least in the near future, the industry is bound to continue growing.

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